

# Design Rules Summary

for Drawing Layers

2026/06/09

Reflected GitHub issues

<https://github.com/OpenSUSI/TR-1um/issues>

# Drawing Layer Table

**\*IMPORTANT\***

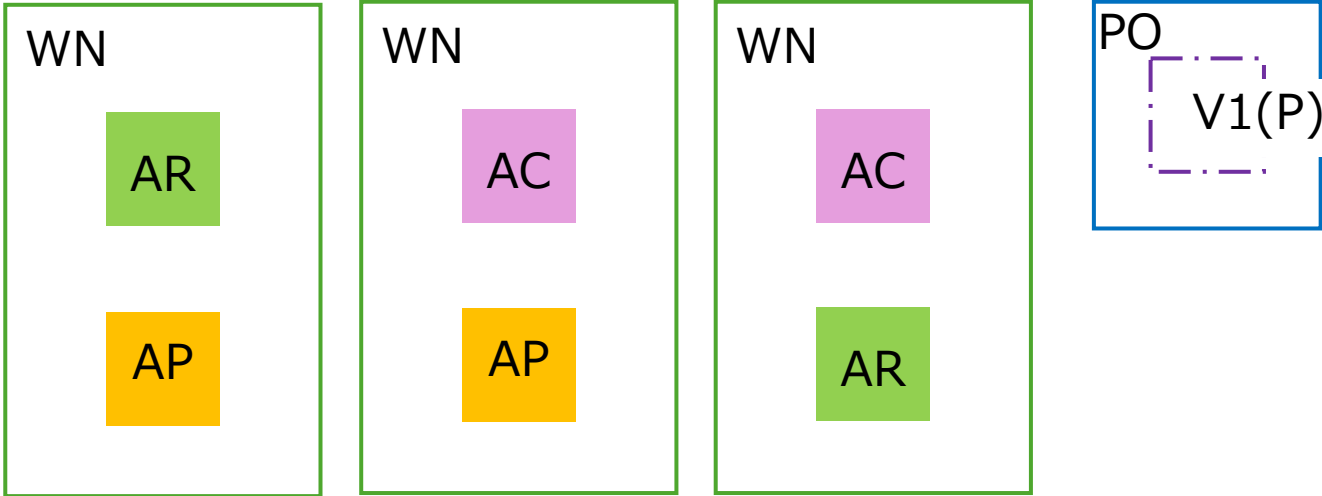
DBU: 0.001um

Scale Factor: 1.0

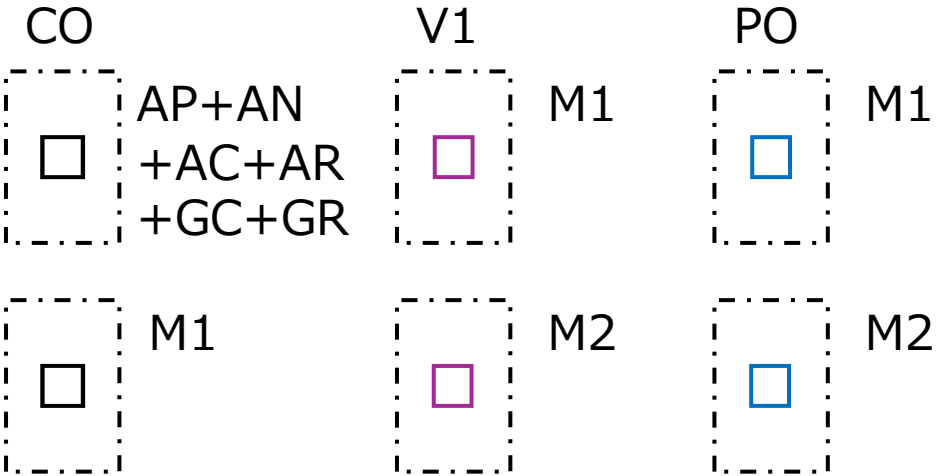
Min Grid: 0.05um

|    | Original MASK layers |     |      |  | Proposed Drawing layers |     |      |                           |  |  |
|----|----------------------|-----|------|--|-------------------------|-----|------|---------------------------|--|--|
|    | Name                 | NUM | TYPE |  | Name                    | NUM | TYPE |                           |  |  |
| 1  | PSUB                 | 140 | 0    |  | WN                      | 140 | 0    |                           |  |  |
| 2  | NW                   | 36  | 0    |  |                         |     |      | MDP generatioed           |  |  |
| 3  | HVNW                 | 141 | 0    |  |                         |     |      | MDP generatioed           |  |  |
| 4  | L                    | 3   | 0    |  |                         |     |      | MDP generatioed           |  |  |
|    |                      |     |      |  | AP                      | 3   | 1    | Active for P+             |  |  |
|    |                      |     |      |  | AN                      | 3   | 2    | Active for N+             |  |  |
|    |                      |     |      |  | AR                      | 3   | 3    | Active for resistor       |  |  |
|    |                      |     |      |  | AC                      | 3   | 4    | Active for capacitance    |  |  |
| 5  | NF                   | 25  | 0    |  |                         |     |      | MDP generatioed           |  |  |
| 6  | PF                   | 26  | 0    |  |                         |     |      | MDP generatioed           |  |  |
| 7  | CL                   | 143 | 0    |  |                         |     |      | MDP generatioed           |  |  |
| 8  | HPBE                 | 144 | 0    |  |                         |     |      | No USE                    |  |  |
| 9  | HNBE                 | 145 | 0    |  |                         |     |      | No USE                    |  |  |
| 10 | PBE                  | 146 | 0    |  |                         |     |      | MDP generatioed           |  |  |
| 11 | NBE                  | 147 | 0    |  |                         |     |      | MDP generatioed           |  |  |
| 12 | HPM                  | 33  | 0    |  |                         |     |      | MDP generatioed           |  |  |
| 13 | RHP                  | 148 | 0    |  |                         |     |      | No USE                    |  |  |
| 14 | SG                   | 8   | 0    |  |                         |     |      |                           |  |  |
|    |                      |     |      |  | GC                      | 8   | 1    | Gate Poly transistor      |  |  |
|    |                      |     |      |  | GR                      | 8   | 2    | Gate POLY resistor        |  |  |
| 15 | PM                   | 35  | 0    |  |                         |     |      | MDP generatioed           |  |  |
| 16 | NM                   | 7   | 0    |  |                         |     |      | MDP generatioed           |  |  |
| 17 | R                    | 12  | 0    |  |                         |     |      | MDP generatioed           |  |  |
| 18 | PSD                  | 9   | 0    |  |                         |     |      | MDP generatioed           |  |  |
| 19 | NSD                  | 28  | 0    |  |                         |     |      | MDP generatioed           |  |  |
| 20 | CONT                 | 11  | 0    |  | CO                      | 11  | 0    |                           |  |  |
| 21 | M1                   | 13  | 0    |  | M1                      | 13  | 0    |                           |  |  |
| 22 | TC                   | 19  | 0    |  | V1                      | 19  | 0    |                           |  |  |
| 23 | M2                   | 20  | 0    |  | M2                      | 20  | 0    |                           |  |  |
| 24 | PRO                  | 14  | 0    |  | PO                      | 14  | 0    |                           |  |  |
|    |                      |     |      |  |                         |     |      |                           |  |  |
|    | TXM1                 | 48  | 0    |  | TXM1                    | 48  | 0    | M1 Label                  |  |  |
|    | PIN                  | 48  | 1    |  | PIN                     | 48  | 1    | M1 Pin                    |  |  |
|    | TXM2                 | 49  | 0    |  | TXM2                    | 49  | 0    | M2 Label                  |  |  |
|    | MASK                 | 63  | 0    |  | MASK                    | 63  | 0    | Waiver region             |  |  |
|    | PTECT                | 63  | 1    |  | PTECT                   | 63  | 1    | Protect region (LOGO)     |  |  |
|    | SCRB                 | 80  | 0    |  | SCRB                    | 80  | 0    | Waiver region (EDGE SEAL) |  |  |
|    | prBoundary           | 235 | 0    |  | prBoundary              | 235 | 0    | Cell boundary             |  |  |

# Cat-1: Prohibited



| Rule  | L1    | L2       | Func  | MIN | MAX |
|-------|-------|----------|-------|-----|-----|
| WN.Z1 | WN(R) | AP       | None  | 0.0 |     |
| WN.Z2 | WN(C) | AP       | None  | 0.0 |     |
| WN.Z3 | WN(C) | AR       | None  | 0.0 |     |
| CO.Z1 | CO    | AA+GC+GR | None  | 0.0 |     |
| CO.Z3 | CO    | M1       | None  | 0.0 |     |
| V1.Z1 | V1    | M1       | None  | 0.0 |     |
| V1.Z2 | V1    | M2       | None  | 0.0 |     |
| PO.Z1 | PO    | M1       | None  | 0.0 |     |
| PO.Z2 | PO    | M2       | None  | 0.0 |     |
| PO.Z3 | PO    | V1(P)    | Exist | 0.0 |     |



**Function:**

Wmin/max: L1 Width min/max

Smin/max: L1 Space min/max (L1=L2)

Smin/max: L1 Separate L2 min/max

Emin/max: L1 Enclosed L2 min/max

Fmin/max: L1 Enclosing L2 min/max

ECmin: MOS Endcap min

Wfix: L1 Windth equal

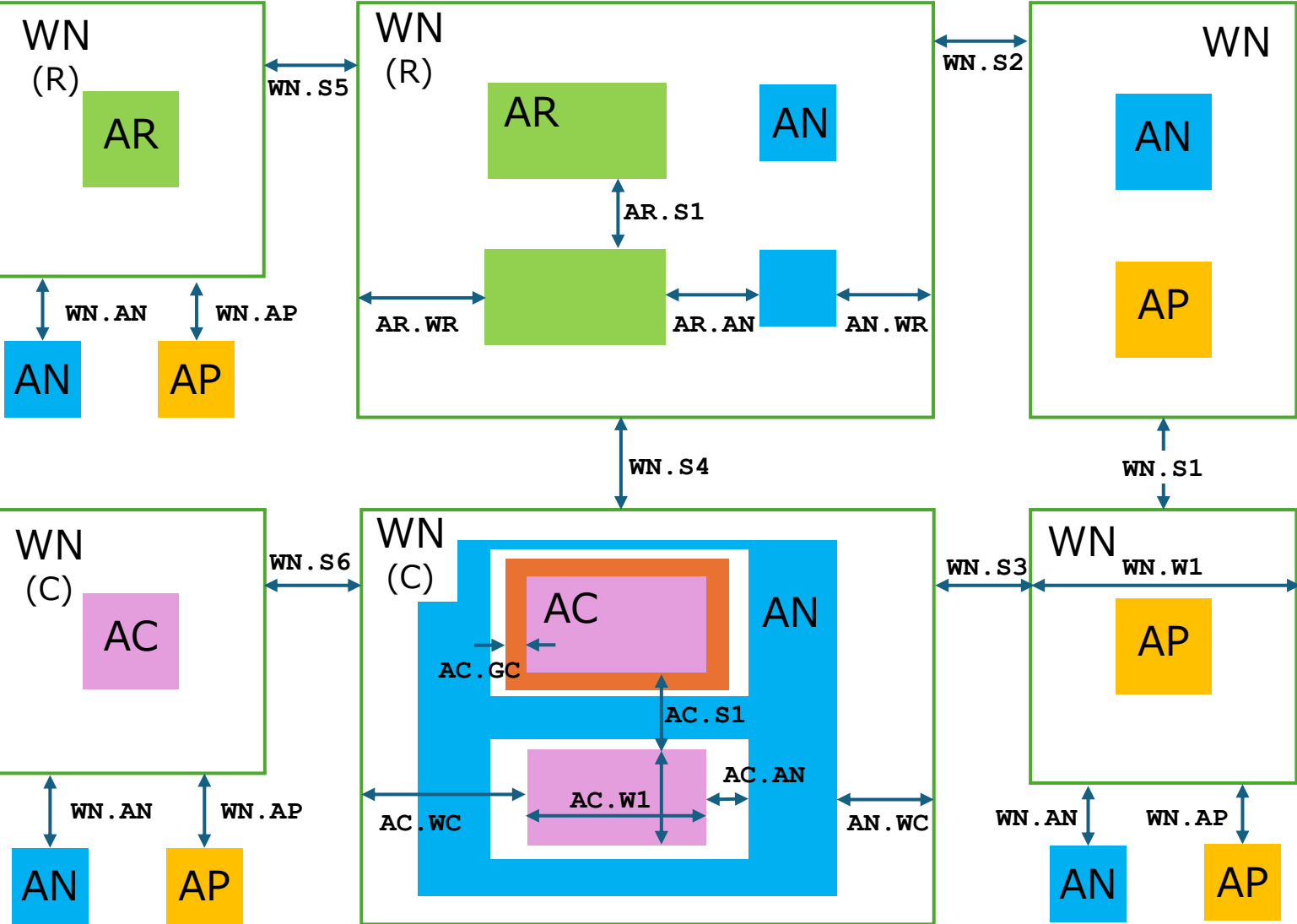
Sfix: Space equal

Donut: L1 Souuounded by L2

None: Not Exist

V1(P) : V1 on PE

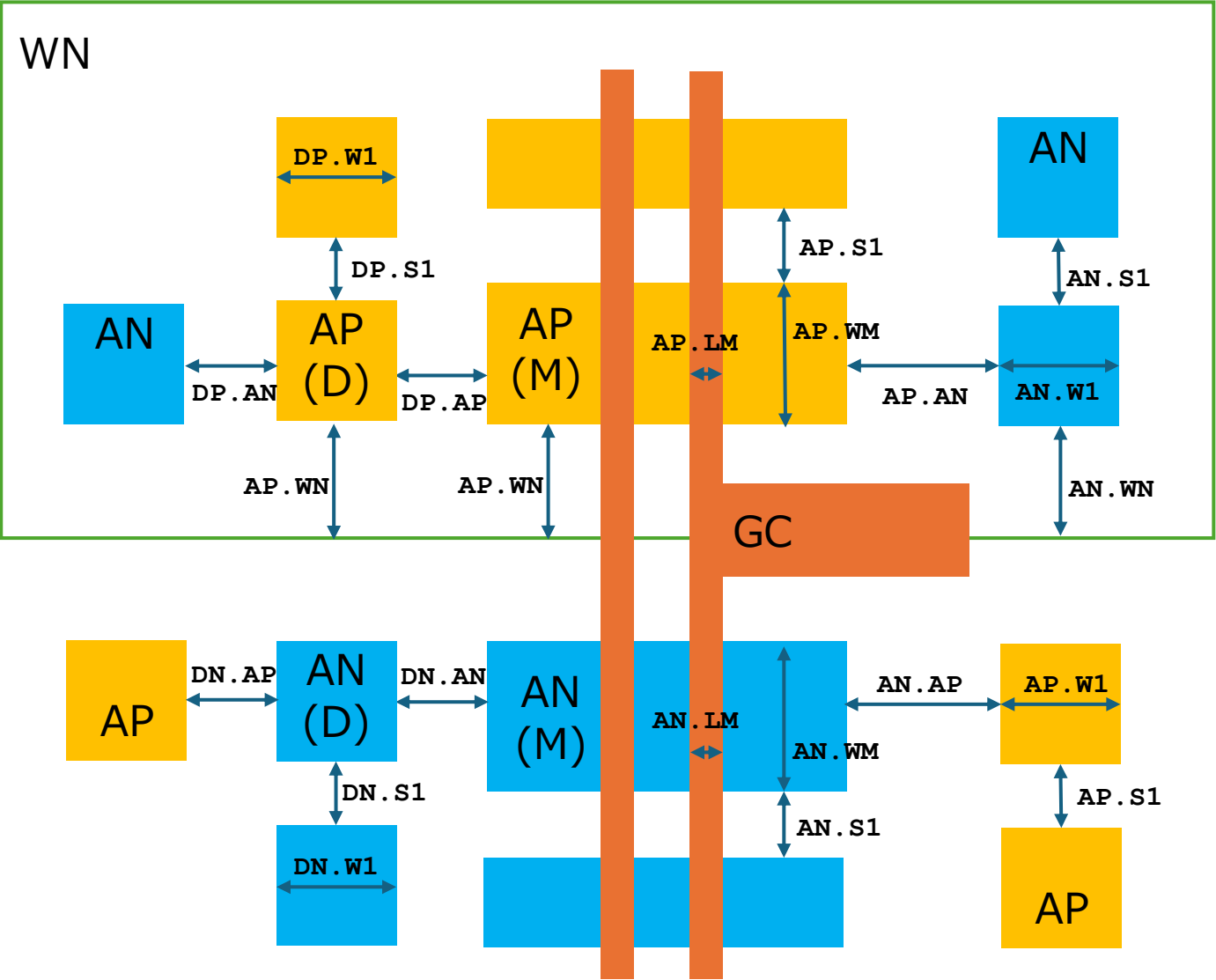
# Cat-2: WN to AP/AN/AR/AC related



| Rule  | L1    | L2    | Func     | MIN  | MAX   |
|-------|-------|-------|----------|------|-------|
| WN.W1 | WN    |       | Wmin     | 8.0  |       |
| WN.S1 | WN    | WN    | Smin     | 12.0 |       |
| WN.S2 | WN    | WN(R) | Smin     | 9.5  |       |
| WN.S3 | WN    | WN(C) | Smin     | 12.0 |       |
| WN.S4 | WN(R) | WN(C) | Smin     | 9.5  |       |
| WN.S5 | WN(R) | WN(R) | Smin     | 8.0  |       |
| WN.S6 | WN(C) | WN(C) | Smin     | 12.0 |       |
| WN.AP | WN    | AP    | Smin     | 5.0  |       |
| WN.AN | WN    | AN    | Smin     | 10.0 |       |
| AR.S1 | AR    | AR    | Smin     | 4.0  |       |
| AR.WR | AR    | WN(R) | Emin     | 10.0 |       |
| AN.WR | AN    | WN(R) | Emin     | 5.0  |       |
| AR.AN | AR    | AN    | Smin     | 4.0  |       |
| AC.W1 | AC    |       | Wmin/max | 28.5 | 120.0 |
| AC.S1 | AC    | AC    | Smin     | 6.4  |       |
| AC.WC | AC    | WN(C) | Emin     | 8.0  |       |
| AN.WC | AN    | WN(C) | Emin     | 3.0  |       |
| AC.AN | AC    | AN    | Sfix     | 2.8  | 2.8   |
| AC.GC | AC    | GC    | Efix     | 1.4  | 1.4   |
| AC.R1 | AC    | AN(C) | Donut    | 0.0  |       |

WN(R) : WN overlapped AR  
WN(C) : WN overlapped AC  
AN(C) : AN in WN(C)

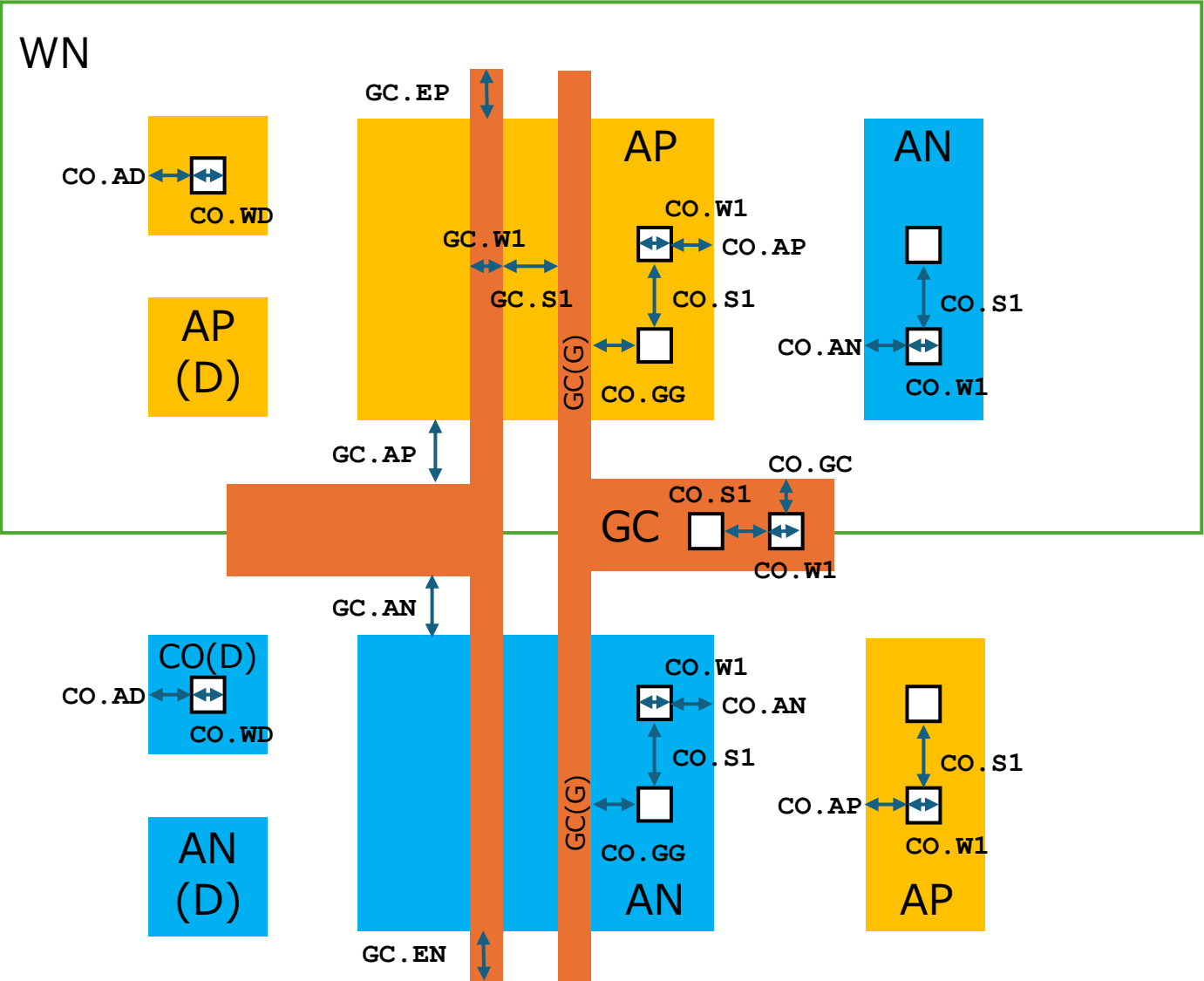
# Cat-3: MOS transistor related



| Rule  | L1   | L2    | Func     | MIN | MAX  |
|-------|------|-------|----------|-----|------|
| AP.W1 | AP   |       | Wmin     | 1.4 |      |
| AP.S1 | AP   | AP    | Smin     | 1.4 |      |
| AP.AN | AP   | AN    | Smin     | 2.8 |      |
| AP.WM | PMOS | AP-GC | Wmin/max | 3.4 | 60.0 |
| AP.LM | PMOS | GC-AN | Lmin/max | 1.0 | 30.0 |
| AP.WN | AP   | WN    | Emin     | 7.0 |      |
| DP.W1 | DP   |       | Wmin     | 2.8 |      |
| DP.S1 | DP   | DP    | Smin     | 1.4 |      |
| DP.AP | DP   | AP(M) | Smin     | 2.8 |      |
| DP.AN | DP   | AN    | Smin     | 2.8 |      |
| AN.W1 | AN   |       | Wmin     | 1.4 |      |
| AN.S1 | AN   | AN    | Smin     | 1.4 |      |
| AN.AP | AN   | AP    | Smin     | 2.8 |      |
| AN.WM | NMOS | AN-GC | Wmin/max | 3.4 | 60.0 |
| AN.LM | NMOS | GC-AN | Lmin/max | 1.0 | 30.0 |
| AN.WN | AN   | WN(M) | Emin     | 5.0 |      |
| DN.W1 | DN   |       | Wmin     | 2.8 |      |
| DN.S1 | DN   | DN    | Smin     | 1.4 |      |
| DN.AP | DN   | AP    | Smin     | 2.8 |      |
| DN.AN | DN   | AN(M) | Smin     | 2.8 |      |

AN (M) : AN-DN  
AP (M) : AP-DP  
WN (M) : WN overlapped AN/AP

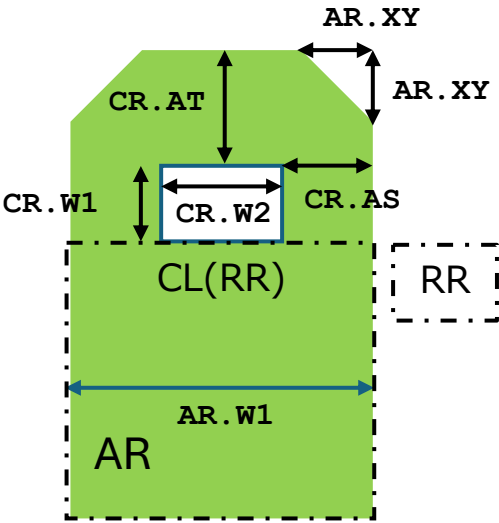
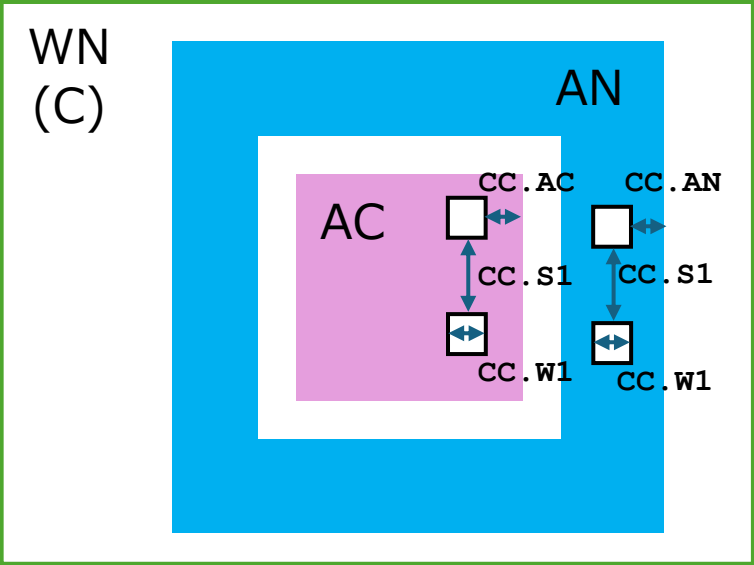
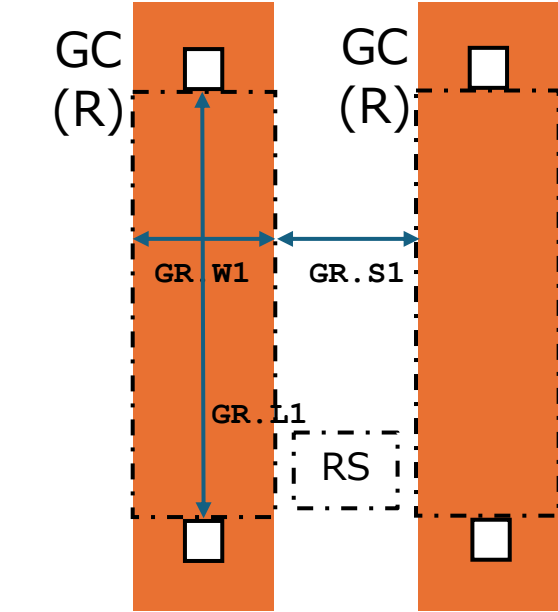
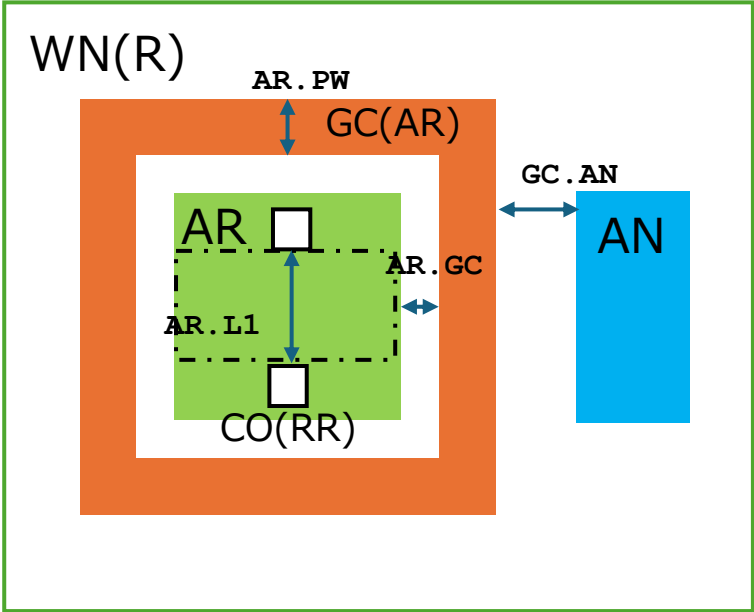
# Cat-4: Contact and Poly related



| Rule  | L1    | L2    | Func  | MIN | MAX |
|-------|-------|-------|-------|-----|-----|
| GC.W1 | GC    |       | Wmin  | 1.0 |     |
| GC.S1 | GC    | GC    | Smin  | 1.2 |     |
| GC.AP | GC    | AP    | Smin  | 0.4 |     |
| GC.AN | GC    | AN    | Smin  | 0.4 |     |
| GC.EP | PMOS  | GC    | ECmin | 1.2 |     |
| GC.EN | NMOS  | GC    | ECmin | 1.2 |     |
| CO.W1 | CO    |       | Wfix  | 1.0 | 1.0 |
| CO.S1 | CO    | CO    | Smin  | 1.0 |     |
| CO.WD | CO(D) |       | Wfix  | 1.2 | 1.2 |
| CO.AD | CO(D) | AP+AN | Emin  | 1.2 |     |
| CO.AP | CO    | AP    | Emin  | 0.8 |     |
| CO.AN | CO    | AN    | Emin  | 0.8 |     |
| CO.GG | CO    | GC(G) | Smin  | 1.0 |     |
| CO.GC | CO    | GC+GR | Emin  | 0.8 |     |

AP(D) : AP not interacted GC  
AN(D) : AN not interacted GC  
CO(D) : CO on AP(D) /AN(D)  
GC(G) : GC for PMOS/NMOS

Cat-5: Resistor(AR/RS) and Capacitor(AC) related

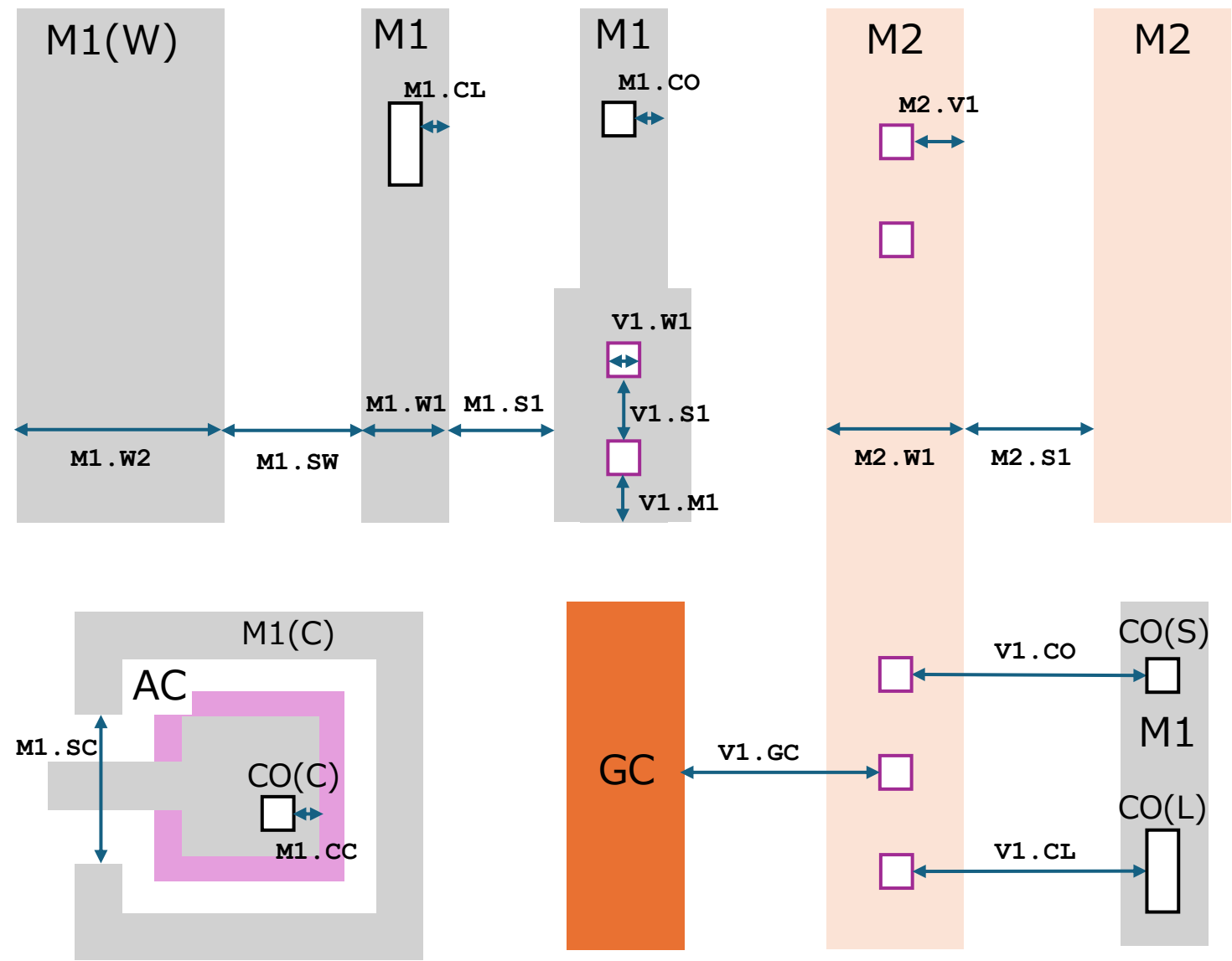


R.Effective region

| Rule  | L1     | L2    | Func     | MIN  | MAX   |
|-------|--------|-------|----------|------|-------|
| AR.W1 | RR     | RR(L) | Wmin/max | 2.8  | 20.0  |
| AR.L1 | RR     | RR(W) | Lmin/max | 13.0 | 100.0 |
| AR.GC | GC     | AR    | Smin     | 1.0  |       |
| AR.PW | GC(R)  |       | Wmin     | 2.0  |       |
| AR.XY | AR     | Bevel | XYmin    | 1.0  |       |
| GC.AN | GC(R)  | AN    | Smin     | ???  | ???   |
| GC.R1 | AR     | GC(R) | Donut    | 0.0  |       |
| GC.R2 | GC(R)  | WN(R) | TieDown  | 0.0  |       |
| GR.W1 | RS     | RS(L) | Wmin/max | 4.0  | 20.0  |
| GR.L1 | RS     | RS(W) | Lmin/max | 20.0 | 100.0 |
| GR.S1 | RS     | RS    | Smin     | 2.0  |       |
| CR.W1 | CO(RR) |       | Wmin     | 1.0  |       |
| CR.W2 | CL(RR) |       | Wmin/max | 1.0  | 97.5  |
| CR.AT | CO(RR) | AR(T) | Emin     | 1.5  |       |
| CR.AS | CO(RR) | AR(S) | Emin     | 1.0  |       |
| CC.W1 | CO(C)  |       | Wfix     | 1.2  | 1.2   |
| CC.S1 | CO(C)  | CO(C) | Sfix     | 1.2  | 1.2   |
| CC.AC | CO(C)  | AC    | Emin     | 2.9  |       |
| CC.AN | CO(C)  | AN    | Emin     | 1.2  |       |

GC(AR) : GC in WN(R)  
RS(L) : Length of AR effective  
RS(W) : Width of AR effective  
CO(RR) : CO short side on AR  
CL(RR) : CO large side on AR  
AR(T) : AR top fringe  
AR(S) : AR side fringe  
AR.L1 : allow only 4,6,12,20  
CO(C) : CO on AC

# Cat-6: M1 /V1 /M2 related

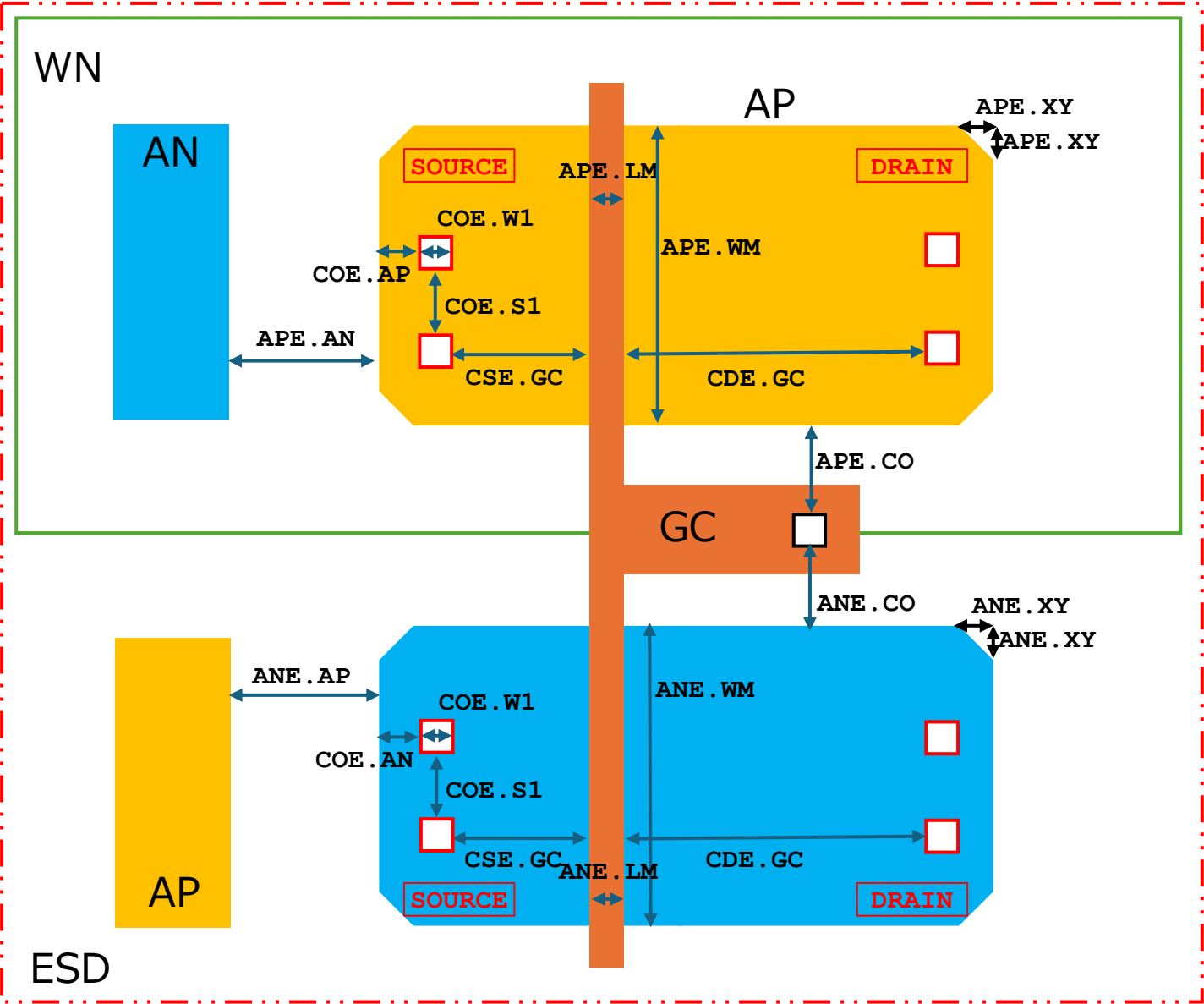


| Rule  | L1    | L2    | Func     | MIN  | MAX  |
|-------|-------|-------|----------|------|------|
| M1.W1 | M1    |       | Wmin/max | 1.8  | 10.0 |
| M1.W2 | M1(W) |       | Wmin     | 10.0 |      |
| M1.S1 | M1    | M1    | Smin     | 1.4  |      |
| M1.SC | M1(C) | M1(C) | Sfix     | 10.0 | 10.0 |
| M1.CO | M1    | CO(S) | Fmin     | 0.8  |      |
| M1.CC | M1    | CO(C) | Fmin     | 1.3  |      |
| M1.CL | M1    | CO(L) | Fmin     | 1.2  |      |
| M1.SW | M1(W) | M1    | Smin     | 2.0  |      |
| V1.W1 | V1    |       | Wfix     | 1.4  | 1.4  |
| V1.S1 | V1    | V1    | Smin     | 1.5  |      |
| V1.M1 | V1    | M1    | Emin     | 1.0  |      |
| V1.GC | V1    | GC    | Smin     | 1.2  |      |
| V1.CO | V1    | CO    | Smin     | 1.0  |      |
| V1.CL | V1    | CO(L) | Smin     | 1.4  |      |
| M2.W1 | M2    |       | Wmin     | 3.0  |      |
| M2.S1 | M2    | M2    | Smin     | 2.0  |      |
| M2.V1 | M2    | V1    | Fmin     | 1.0  |      |

M1 (C) : M1 in WN (C)  
CO (C) : CO on AC  
CO (S) : CO small contact  
CO (L) : CO large contact (only on AR)  
M1 (W) : Wide M1



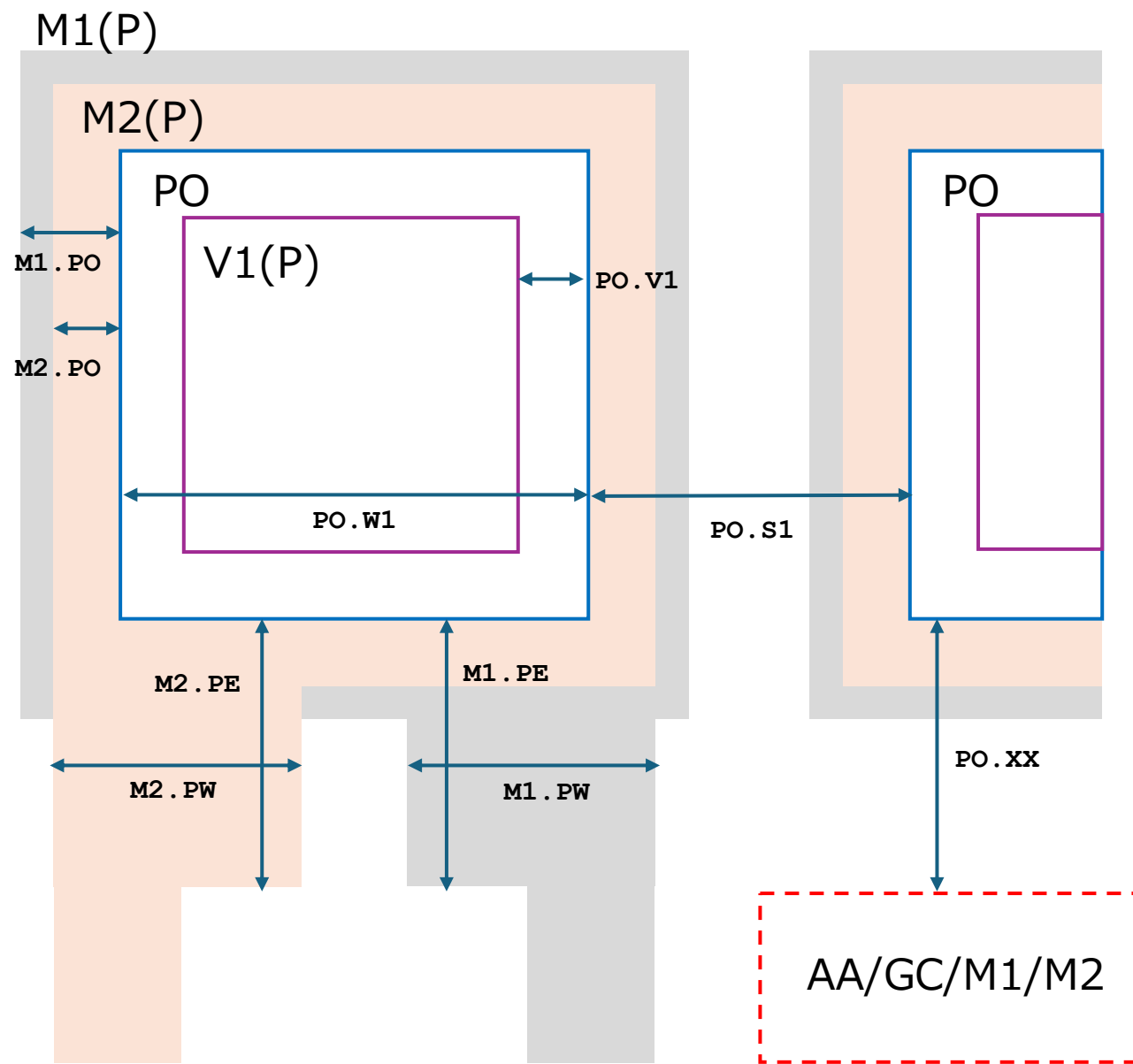
# Cat-7: ESD MOS transistor related



| Rule   | L1    | L2     | Func     | MIN  | MAX  |
|--------|-------|--------|----------|------|------|
| APE.WM | PMOSE | APE-GC | Wmin/max | 11.0 | 60.0 |
| APE.LM | PMOSE | GC-ANE | Lmin/max | 2.0  | 30.0 |
| APE.AN | APE   | AN     | Smin     | 10.0 |      |
| APE.XY | APE   | Bevel  | XYmin    | 1.4  |      |
| APE.CO | APE   | CO     | Smin     | 2.0  |      |
| ANE.WM | NMOSE | ANE-GC | Wmin/max | 11.0 | 60.0 |
| ANE.LM | NMOSE | GC-ANE | Lmin/max | 2.0  | 30.0 |
| ANE.AP | ANE   | AP     | Smin     | 10.0 |      |
| ANE.XY | ANE   | Bevel  | XYmin    | 1.4  |      |
| ANE.CO | ANE   | CO     | Smin     | 2.0  |      |
| COE.W1 | CO(E) |        | Wfix     | 3.0  | 3.0  |
| COE.S1 | CO(E) | CO(E)  | Smin     | 1.6  |      |
| CDE.GC | CD(E) | GC     | Smin     | 7.0  |      |
| CSE.GC | CS(E) | GC     | Smin     | 3.0  |      |
| COE.AP | CO(E) | AP     | Emin     | 4.0  |      |
| COE.AN | CO(E) | AN     | Emin     | 4.0  |      |

APE/ANE) : AP/AN on ESD)  
CO (E) : CO on ESD

# Cat-8: PAD related



| Rule  | L1     | L2        | Func | MIN  | MAX |
|-------|--------|-----------|------|------|-----|
| PO.W1 | PO     |           | Wmin | 70.0 |     |
| PO.S1 | PO     |           | Smin | 64.0 |     |
| M2.PO | M2 (P) | PO        | Fmin | 5.0  |     |
| M1.PO | M1 (P) | PO        | Fmin | 5.0  |     |
| PO.V1 | PO     | V1 (P)    | Emin | 10.0 |     |
| PO.AP | PO     | AP        | Smin | 14.0 |     |
| PO.AN | PO     | AN        | Smin | 14.0 |     |
| PO.GC | PO     | GC        | Smin | 14.0 |     |
| PO.M1 | PO     | M1        | Smin | 14.0 |     |
| PO.M2 | PO     | M2        | Smin | 14.0 |     |
| M2.PE | M2     | Extention |      | 19.0 |     |
| M2.PW | M2     | Extention | Wmin | 40.0 |     |
| M1.PE | M2     | Extention |      | 19.0 |     |
| M1.PW | M2     | Extention | Wmin | 40.0 |     |

\*\* Not yet implemented

M1 (P) : M1 on PO  
M2 (P) : M2 on PO  
V1 (P) : V1 on PO

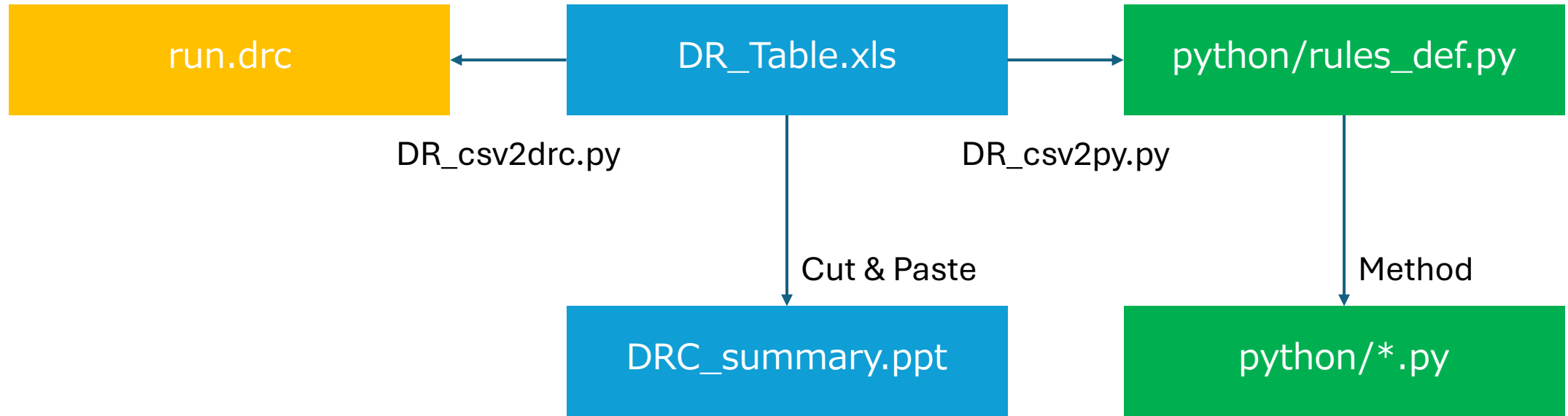
# Design Rule control

DRC integrity

# DRC

# Document

# PCell



# ALL Critical Dimension parameters are referenced by 'XX.YY' rule's name.

# Design to Tapeout Process

Mask Data Preparation

# Designer

local



[push]

remote



[action]  
Precheck  
DRC  
MDP



[clone]



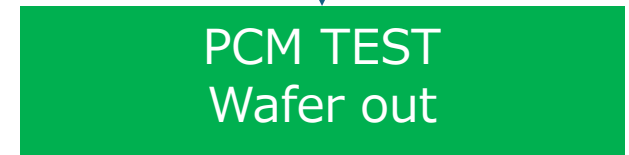
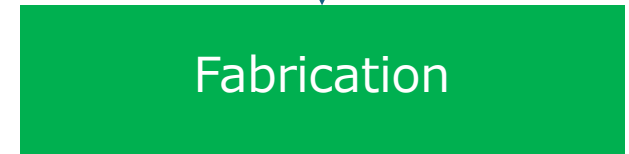
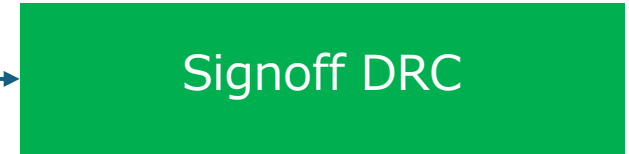
[Import artifact]



[action]  
Logo insertion  
Aggregation  
DRC



# Tokai Rika



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